159302 – Artificial Intelligence
Tutorial #10 - Propositional Logic

The following notations are used in the questions that follow:

- **^** (caret) for and,
- **v** (lower case v) for or,
- **~** for not,
- **->** (hyphen, >) for implication, and
- **<->** (<, hyphen, >) for biconditionals.
- Propositional variables are upper case letters (and possibly numbers).
- The symbols true and false are written out in lower case.

¬ ∧ ∨ → ↔

I. Precedence

Give a fully parenthesized version of each of these expressions. Each of your answers needs to follow the definitions legal sentences in propositional logic. Preserve the original order of the variable names in your parenthesized expression.

1. (P v (~ Q)) -> R
2. (P ^ Q) v R
3. (P -> Q) <-> (R -> S)
4. ((~ R) v S) -> (Q ^ R)

II. Translating Sentences

If we let the propositional variables S, Z, R, and H have the following intended meanings:

- S = the sun shines
- Z = we go to the zoo
- R = it rains
- H = we stay at home

Give, for each of the following English sentences, a propositional logic sentence with the same meaning:

1. If the sun shines then we go to the zoo.
2. It rains or the sun shines.
3. If we do not go to the zoo then we stay at home.
4. If we do not go to the zoo then it rains.
5. If we do not stay at home then the sun shines.
6. If the sun shines or we do not stay at home then we go to the zoo.
7. If the sun does not shine then we do not go to the zoo.

III. Simplifications

Use the semantics rules to simplify the following formulas as much as possible.

1. P v true — true
2. P ^ true — P
3. P v false — P
4. P ^ false — false
5. (P ^ false) v Q — Q
6. (P ^ Q) v true — true
7. P ^ (Q v true) — P
8. P ^ (Q v false) — P ^ Q

II

1. $S \rightarrow Z$
2. $(R \vee S)$
3. $\neg Z \rightarrow H$
4. $\neg Z \rightarrow R$
5. $\neg H \rightarrow S$
6. $(S \vee \neg H) \rightarrow Z$
7. $\neg S \rightarrow \neg Z$

IV CNF

1. $(A \wedge \neg B) \rightarrow C \equiv \neg(A \wedge \neg B) \vee C$

$\equiv (\neg A \vee B) \vee C$

$\equiv (\neg A \vee B \vee C)$ ✓

2. $(A \rightarrow B) \rightarrow C \equiv (\neg A \vee B) \rightarrow C$

$\equiv \neg(\neg A \vee B) \vee C$

$\equiv (A \wedge \neg B) \vee C$

$\equiv (A \vee C) \wedge (\neg B \vee C)$ ✓

3. $A \rightarrow (B \rightarrow C) \equiv \neg A \vee (B \rightarrow C)$

$\equiv \neg A \vee (\neg B \vee C)$

$= (\neg A \vee \neg B \vee C)$ ✓

IV CNF

4. $((A \wedge B) \vee C) \rightarrow D \equiv \neg((A \wedge B) \vee C) \vee D$

$\equiv ((\neg A \vee \neg B) \wedge \neg C) \vee D$

another

$\equiv (\neg A \vee \neg B \vee D) \wedge (\neg C \vee D)$

5. $((A \wedge B) \vee C) \rightarrow (D \wedge E) \equiv$

$\equiv \neg((A \wedge B) \vee C) \vee (D \wedge E)$

$\equiv ((\neg A \vee \neg B) \wedge \neg C) \vee (D \wedge E)$

$\equiv (\neg A \vee \neg B \vee D) \wedge (\neg C \vee D)$

$\wedge (\neg A \vee \neg B \vee E) \wedge (\neg C \vee E)$ ✓

$\equiv$

立. Propositional Resolution Refutation

1. $\neg P \vee Q \vee S$
2. $\neg S$
3. $\neg Q \vee X$
4. $\neg X$
5. P —— negated conclusion

$$\frac{\alpha \vee \beta \qquad \neg\beta \vee \gamma}{\alpha \vee \gamma}$$

SOLUTION #1

6.	$Q \vee S$	1, 5
7.	$\neg P \vee Q$	1, 2
8.	$\neg Q$	3, 4
9.	$\neg P$	7, 8
10	■	5, 9

SOLUTION #2

6.	$\neg P \vee Q$	(1, 2)
7.	$\neg Q$	(3, 4)
8.	$\neg P$	(6, 7)
9.	■	(5, 8)